Summer Internship Presentation

Real-Time Chat & Authentication Systems

Presented By

Harsh Gharewal

Academic Credentials

Branch: Mechanical Engineering

College: SGSITS College

Enrollment No.: 0801ME231043

Host Institute

Indian Institute of Technology Goa

Internship Duration

01/04/2026 – 10/06/2026

Official Internship Certificate

भारतीय प्रौद्योगिकी संस्थान गोवा
गोवा अभियांत्रिकी महाविद्यालय परिसर, फार्मागुड़ी, फ़ोंडा – 403401, गोवा

Indian Institute of Technology Goa
Goa College of Engineering Campus, Farmagudi, Ponda – 403401, Goa



IIT Goa

Date:22 September 2026

TO WHOMEVER IT MAY CONCERN

This is to certify that Mr. Harsh Gharewal, a final-year student at Shri Govindram Seksaria Institute of Technology and Science (SGSITS), Indore, pursuing B.Tech in Mechanical Engineering (Roll No. 0801ME231043), worked as a remote intern with me.

The internship was conducted from 1 April 2026 to 10 June 2026, during which he engaged in project-oriented work and contributed to the assigned project activities. His conduct and performance during the internship were satisfactory.

I wish him the very best in all his future academic and professional endeavors.



Dr. Rahul CS
School of Mathematics & Computer Science
Indian Institute of Technology Goa
India

IIT Goa Strives to Address Global Challenges Through Advanced Science and Technology Education

VISION

To become a unique institution of higher learning, offering state-of-the-art education, research, and training in science and technology to address societal and global challenges.

MISSION

Empowering students as responsible leaders, managers, and innovators for sustainable development.

Short-Term Objectives

- Delivering high-quality education
- Advanced multidisciplinary research

Long-Term Goals

- Achieving self-sustainability
- Translational research focus
- Global institutional partnerships

FACULTY MENTORSHIP & ADVISORY

Structured Mentorship

Facilitated Progressive Problem-Definition and Architectural Design

MENTOR DETAILS

Dr. Rahul C. S.

Associate Professor

Department of Computer Science & Engineering

IIT Goa

Provided continuous technical leadership, structured feedback, and strategic architectural direction throughout the conceptualization process.

MENTORSHIP IMPACT & METHODOLOGY

Progressive Problem-Definition

Guided walkthroughs establishing rigorous problem boundaries, refining baseline requirements, and structuring technical scope.

Structured Architectural Guidance

Expert advisory on high-level system design, modular software architecture, and sustainable engineering principles.

Collaborative Problem Breakdown

Enabled effective decomposition of complex engineering challenges, driving clear conceptualization of user requirements and overarching system goals.

System Architecture Integration

Full-Stack MERN Architecture with Socket.io Real-Time Synchronization

mongodb | express.js | react.js | node.js | **socket.io**

Real-Time Engine

Multi-User Chat & Sync

Integrates Socket.io for continuous, bidirectional real-time data synchronization across active user sessions.

Persistent Data Storage

Backed by MongoDB to provide reliable, scalable storage for chat history, user profiles, and operational logs.

Multi-Factor Security

Two-Factor Authentication

Integrated multi-factor security verification layer securing user access across REST and WebSocket services.

Email & SMS Dispatchers

Leverages Nodemailer for automated email OTP generation and Twilio API integration for SMS-based verification.

State Synchronization

Unified Architecture

Ensures synchronized state management seamlessly maintained between the React.js frontend interface and backend services.

Dual Protocol Backend

Combines secure REST API endpoints via Express.js with low-latency WebSocket communication pipelines.

Continuous Milestone Tracking & Iterative Sprints

Ensuring Strict Architectural Alignment Across Development Cycles

METHODOLOGY OVERVIEW

Structured Management

Project management utilized structured review cycles held every weekend to audit completed deliverables.

Iterative feedback loops kept all components in strict alignment with overall system goals.

CORE OUTCOME

100% Architectural Alignment

PHASE 01 // WEEKEND AUDITS

Structured Review Cycles

Weekly audits conducted every weekend to systematically evaluate and verify all completed deliverables.

PHASE 02 // SPRINT EXECUTION

Tracking & Blocker Identification

Continuous milestone tracking coupled with proactive blocker identification and iterative sprint planning.

PHASE 03 // SYSTEM INTEGRATION

Interactive Feedback Loops

Continuous feedback loops established throughout development to maintain strict system architecture integrity.

01 // ENGINEERING MECHANICS

Algorithmic Thinking & Mechanics

The project successfully bridged foundational algorithmic thinking and systems engineering principles with modern full-stack web architectures.

- **Systems Engineering Principles**
  Mathematical rigor & core mechanics
- **Complex Data Models**
  Structured state representations

02 // WEB ARCHITECTURES

Full-Stack Web Systems

This cross-disciplinary synthesis involved the practical application of complex data models, asynchronous programming logic, and distributed client-server state synchronization to build a robust real-time communication platform.

- **Asynchronous Programming Logic**
  Event-driven non-blocking execution
- **Distributed State Synchronization**
  Real-time client-server communication

Key Technical Takeaways

Event-Driven JavaScript & Security Implementation

- **Event-Driven Asynchronous JavaScript & WebSockets:** Mastered building real-time applications leveraging asynchronous workflows and WebSocket protocols.
- **Robust Security Layer Implementation:** Gained hands-on experience in implementing JWT authentication, OTP verification, and API rate-limiting.
- **Scalable Database Design & Production Debugging:** Enhanced practical skills in scalable database schema design and real-world production debugging workflows.

Overcoming Real-Time Communication Latency and Race Conditions Drove Engineering Growth

// engineering_challenges

- **WebSocket State Management:** Handling complex race conditions and unpredictable connection drops in real-time communication flows.
- **Third-Party Latency:** Integrating and debugging third-party API webhooks, such as Twilio and Nodemailer, under strict low-latency constraints.
- **Domain Transition:** Rapidly pivoting from mechanical engineering paradigms to production-grade modern web stacks.

// solutions_&_growth

- **Advanced Troubleshooting:** Implemented robust state handling and reconnection strategies for resilient socket connections.
- **Optimized Webhooks:** Streamlined asynchronous processing pipelines to maintain low latency across external integrations.
- **Technical Adaptability:** Demonstrated high adaptability by rapidly mastering full-stack web development and architecture design.

Project Deliverables & Acknowledgments

Conclusion of Internship & Institutional Gratitude

PRIMARY OUTCOME

The internship concluded with the successful delivery of a secured, scalable real-time messaging application.

Institutional Acknowledgments

GUIDANCE & RESEARCH INFRASTRUCTURE

IIT Goa & CSE Department

Sincere gratitude is extended to Dr. Rahul C. S., the Department of Computer Science & Engineering, and IIT Goa for their invaluable guidance and research infrastructure.

ACADEMIC & INSTITUTIONAL SUPPORT

SGSITS College

Deep appreciation is also directed to SGSITS College for providing institutional encouragement and academic support.

Thank You